

Hadi Mohammadi

Senior AI & Data Science Expert · PhD in Explainable NLP

📍 Utrecht, The Netherlands ✉ h.mohammadi@uu.nl 🌐 mohammadi.cv 🔗 mohammadi-hadi
 🏠 mohammadi-hadi 📞 0000-0003-0860-9200 📄 Google Scholar

Profile

Researcher in Explainable Natural Language Processing for social-science applications, having completed my PhD in the Department of Methodology & Statistics at Utrecht University. Across six doctoral research projects, I develop transparent and reliable methods for large language models, covering LLM-as-judge evaluation, annotation quality, cultural alignment, and token-level robustness, and connect them to applied data science questions.

I work as an independent researcher who leads projects end to end — alongside the PhD I have led five competitively funded projects, taking each from proposal through to delivery. I now serve as Senior AI & Data Science Expert at AcademicTransfer, where I lead applied LLM work in production and am technical lead of three applied AI projects. I am committed to open, reproducible research, releasing code and data with my work.

Education

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| <p>PhD Utrecht University, Explainable Natural Language Processing for Social Sciences</p> <ul style="list-style-type: none"> • Faculty of Social and Behavioural Sciences, Department of Methodology & Statistics. • Thesis: <i>Let Me Explain! Explainable NLP for Understanding Large Language Models</i>. • Member of the Natural Language and Text Processing (NLTP) group and the Human Data Science (HDS) Lab. • Supervisors: Dr. Robert A. Bagheri, Dr. Anastasia Giachanou, and Prof. Dr. Daniel Oberski. | <p>Utrecht, The Netherlands
Feb 2023 – May 2026</p> |
| <p>MSc University of Tehran, Industrial Engineering, Macro Systems</p> <ul style="list-style-type: none"> • GPA: 18.96/20 (4/4) — ranked 2nd among 50+ graduate students. • Thesis: dynamic pricing with demand learning using reinforcement learning in a non-stationary environment, considering customer behaviour and advertisement (a data-driven approach). • Cognitive Science Researcher with the Institute for Cognitive Science Studies (ICSS) on customer behaviour in dynamic pricing. | <p>Tehran, Iran
Sept 2018 – Apr 2021</p> |
| <p>BSc Isfahan University of Technology, Golpayegan College of Engineering, Mechanical Engineering</p> | <p>Isfahan, Iran
Sept 2011 – July 2016</p> |

Publications

- | | |
|---|-------------|
| <p>EvalMORAAL: Interpretable Chain-of-Thought and LLM-as-Judge Evaluation for Moral Alignment in Large Language Models
 Mohammadi, H., Giachanou, A., Bagheri, R. A.
 aclanthology.org/2026.starsem-conference.34 (Proceedings of the 15th Joint Conference on Lexical and Computational Semantics (*SEM 2026), ACL, pp. 497–515)</p> | <p>2026</p> |
| <p>Exploring Cultural Variations in Moral Judgments with Large Language Models
 Mohammadi, H., Bagheri, R. A.
 clinjournal.org/clinj/article/view/238 (Computational Linguistics in the Netherlands Journal, Vol. 15, pp. 59–77)</p> | <p>2026</p> |

Evaluating GRPO and DPO for Faithful Chain-of-Thought Reasoning in LLMs Mohammadi, H. , Kozák, T., Giachanou, A. arxiv.org/abs/2512.22631 (arXiv preprint)	2025
Assessing the Reliability of LLM Annotations in the Context of Demographic Bias and Model Explanation Mohammadi, H. [†] , Shahedi, T. [†] , Mosteiro, P., Poesio, M., Bagheri, R. A., Giachanou, A. 10.18653/v1/2025.gebnlp-1.9 (Workshop on Gender Bias in Natural Language Processing (GeBNLP), ACL 2025, pp. 92–104 ([†] equal contribution))	2025
Do Large Language Models Understand Morality Across Cultures? Mohammadi, H. , Meijer, Y. F. S. S., Papadopoulou, E., Bagheri, R. A. aclanthology.org/2025.luhme-1.3 (Proceedings of the 2nd LUHME Workshop, ECAI 2025, pp. 30–39)	2025
Explainability-Based Token Replacement on LLM-Generated Text Mohammadi, H. , Giachanou, A., Oberski, D. L., Bagheri, R. A. arxiv.org/abs/2506.04050 (arXiv preprint)	2025
Explainability in Practice: A Survey of Explainable NLP Across Various Domains Mohammadi, H. , Bagheri, R. A., Giachanou, A., Oberski, D. L. arxiv.org/abs/2502.00837 (arXiv preprint)	2025
Measuring Firm Characteristics with Natural Language Processing (NLP) Models: An Error-Mitigating Framework with Tutorial Guidance Oberski, D. L., Domenech, J., Garcia-Bernardo, J., Gerbrands, P., Hassink, W., Mohammadi, H. , Mustica, P., Schilpzand, R., Witteloostuijn, A. (Strategic Organization (under revision))	2025
Novel Approaches in Financial Fraud Detection: Hybrid Machine Learning and Uncertainty-Based Deep Learning Mohammadi, H. , Rahmati, M., Shahedi, T. (Proceedings of BNAIC/BeNeLearn 2024, Utrecht, The Netherlands)	2024
A Transparent Pipeline for Identifying Sexism in Social Media: Combining Explainability with Model Prediction Mohammadi, H. , Giachanou, A., Bagheri, R. A. 10.3390/app14198620 (Applied Sciences, 14(19), 8620)	2024
The CLIN33 Shared Task on the Detection of Text Generated by Large Language Models Fivez, P., Daelemans, W., Van de Cruys, T., Kashnitsky, Y., Chamezopoulos, S., Mohammadi, H. , Giachanou, A., Bagheri, R. A., Poelman, W., Vladika, J., Ploeger, E., Bjerva, J., Matthes, F., van Halteren, H. clinjournal.org/clinj/article/view/182 (Computational Linguistics in the Netherlands Journal, 13, 233–259)	2024
Towards Robust Online Sexism Detection: A Multi-Model Approach with BERT, XLM-RoBERTa, and DistilBERT for EXIST 2023 Tasks Mohammadi, H. , Giachanou, A., Bagheri, R. A. ceur-ws.org/Vol-3497/paper-085.pdf (Working Notes of CLEF 2023, CEUR-WS Vol-3497, pp. 1000–1011)	2023

Statistical Reinforcement Learning with Application in Dynamic Pricing Problem: Learning from Customer Interaction based on Bayesian Approach

2022

Mohammadi, H.

(Tehran: Arshadan Publication (in Persian))

Home Health Care Problem with Synchronization Visits and Considering Samples Transferring Time: A Case Study in Tehran, Iran

2022

Mirabnejad, M., **Mohammadi, H.**, Mirzabaghi, M., Aghsami, A., Jolai, F., Yazdani, M.

[10.3390/ijerph192215036](#) (International Journal of Environmental Research and Public Health, 19(22), 15036)

Grants & Awards

- **Best Oral Presentation**, IOPS Winter Conference 2025, for work on cultural moral alignment in large language models (December 2025).
- **Applied Data Science Grant, Utrecht University (€5K)** — “Optimizing the CV Priority Sorter for Fair and Efficient Prioritization,” with Dr. Robert A. Bagheri, Dr. Georg Kreml (Utrecht University), and Jeroen Sparla (AcademicTransfer) (December 2025).
- **Applied Data Science Grant, Utrecht University (€5K)** — “Rule Extraction from Human Evaluation Using LLMs and Reinforcement Learning,” with Dr. Anastasia Giachanou and Dr. Shihan Wang (July 2025).
- **Centre for Unusual Collaborations (CUCo) Spark Grant (€9K)** — “ReDOSE: Towards a More Circular Pharmaceutical Industry Using AI,” with Dr. Negin Salimi (WUR), Dr. Shayegheh Ashourizadeh (WUR), Dr. Duru Bayram (TU/e), and Dr. Saeed Arbabi (UMCU) (May 2025).
- **European Lighthouse to Manifest Trustworthy and Green AI (Enfield) grant (€14.4K)** — “User Perspectives on Explainable AI” (December 2024).
- **Applied Data Science Grant, Utrecht University (€7.5K)** — “Assessing Reliability of Annotations in the Context of Model Predictions and Explanations,” with Dr. Pablo Mosteiro Romero, Dr. Anastasia Giachanou, and Prof. Dr. Massimo Poesio (December 2023).
- **Scholarship for admission**, Explainable AI (XAI) Summer School, TU Delft, The Netherlands (August 2022).
- **Ranked 2nd** among 50+ Industrial Engineering graduate students, University of Tehran (April 2021).
- **Best Paper Award**, Second National Conference of the Iranian System Dynamics Society (2019).
- **Ranked 35th** of 10,000+ participants, national Iranian MSc entrance exam in Industrial Engineering (September 2018).

Leadership & Representation

- **Co-founder, Young Scholar Network for NLP@U** — established in November 2024 to unite PhD candidates, postdocs, and junior researchers working on NLP and text mining across Utrecht University and UMC Utrecht.
- **LERU Doctoral Summer School Representative** — selected as the representative of the Faculty of Social and Behavioural Sciences, Utrecht University, for the 2025 LERU Doctoral Summer School.

Teaching Experience

Transformers Workshop, UMC Utrecht , Practical Assistant	Utrecht, The Netherlands
• Supervised practical sessions in Python alongside Dr. Robert A. Bagheri and Prof. Dr. Albert Gatt.	Dec 2023 – Dec 2023
Data Wrangling and Data Analysis, Utrecht University , Computer Lab Assistant	Utrecht, The Netherlands
• Assisted in the MSc Applied Data Science course, focusing on R for data visualisation, exploratory data analysis, supervised learning, and text mining.	Sept 2023 – Oct 2023
Deep Learning Summer School, Neuromatch Academy , Lead Teaching Assistant	Remote
• Facilitated learning for an international student cohort and coordinated with mentors and organisers.	Aug 2021 – Sept 2021

Supervision

- ADS MSc theses 2024 (co-supervised with Dr. Robert A. Bagheri): Mijntje Meijer, Evi Papadopoulou, Lourenço Santos Moitinho de Almeida.
- ADS MSc theses 2025 (co-supervised with Dr. Anastasia Giachanou): Hein Brouwer, Tamás Kozák, Mohamad Jouhar.
- ADS MSc theses 2026 (co-supervised with Dr. Robert A. Bagheri): Aliyah Vos, Geanina Verestiuc, Thomas Rietman.

Selected Work Experience

AcademicTransfer, Senior AI & Data Science Expert

Utrecht, The Netherlands

Dec 2025 – present

- Lead AI strategy and applied data-science delivery for academic hiring, translating research on explainable and reliable NLP into production features.
- Technical lead of a recognition-and-rewards policy analysis examining how promotion and reward policies are reflected in academic positions, with Dr. Sanli Faez (national manager, Recognition & Rewards, Universiteiten van Nederland) and university policy makers across the Netherlands.
- Technical lead of a new AI intelligence layer and applicant tracking system for Academic-Transfer, used by 22 Dutch research universities and university medical centres: advanced AI spanning CV creation and revision, semantic search, talent-pool matching, and best-candidate identification.
- Developing a new e-learning programme with the University of Amsterdam for PhD candidates and researchers.

AcademicTransfer, AI & Data Science Expert

Utrecht, The Netherlands

Jan 2024 – Dec 2025

- Developed the CV Priority Sorter using active learning and large language models, cutting HR hiring time by 20%, in collaboration with Utrecht University, TU Delft, and Maastricht University.
- Built end-to-end generative-AI pipelines for automatic insight extraction and de-identification from CVs, improving candidate-job alignment accuracy by 17% and halving manual effort.
- Deployed transformer-based models on Azure ML with MLflow, Docker, and CI/CD; used A/B tests, uplift analysis, and bandit simulations to evaluate and improve ranking and recommendation models.

Utrecht University, School of Economics, Researcher & NLP Developer, FIRMBACKBONE project

Utrecht, The Netherlands

Feb 2024 – Mar 2024

- Developed fbb-sustainability-analysis-cli, adding a keyword extractor for sustainability terms and an index generator that scores websites on sustainability content.

SnowaTec (Innovation Center) — Entekhab Electronic, Senior Data Scientist, Customer Behaviour

Tehran, Iran

Dec 2021 – Dec 2022

- Managed the enterprise data warehouse with modern ETL tooling, improving data accessibility by 16%.
- Led the project to define, capture, and improve KPIs and a customer-satisfaction index, lifting satisfaction by 2%.
- Created interactive Power BI dashboards to support data-driven decisions at all management levels.
- Implemented ML-based customer segmentation, boosting marketing impact and customer experience by 7.5%.

Bdood.bikes (Shared Bicycle System), Head of Data Science & Business Intelligence Team

Tehran, Iran

Apr 2020 – Oct 2020

- Led the data science and business intelligence team and its analytics roadmap for the bike-sharing operation.
- Implemented dynamic pricing and ML-based usage predictions, increasing revenue and rider experience by 5%.
- Built dashboards with Power BI, Streamlit, and Dash Plotly, boosting engagement by 3%, efficiency by 7%, and station space utilisation by 11%.
- Optimised fleet intelligence, cutting maintenance and relocation costs by 8% and 3% respectively, and raising system performance by 2.7%.
- Led user-behaviour analyses that improved customer retention by 4.5%.

Bdood.bikes (Shared Bicycle System), Data Scientist

Tehran, Iran

July 2019 – Apr 2020

- Led predictive-maintenance analysis and battery-health algorithm development, reducing downtime by 3.8% and extending battery life.
- Applied GIS models to identify high-risk riders, enhancing safety by 3% and user experience by 5%.
- Provided marketing and business-analytics insights that contributed to a 6% rise in revenue and engagement.

Additional Professional Development

Generative Modeling Summer School, Eindhoven University of Technology

Eindhoven, The Netherlands

Focus on deep generative modeling.

June 2024

Machine Learning Summer School, University of Oxford

Oxford, UK

Specialisation in finance & NLP/health.

May – July 2023

Explainable AI (XAI) Summer School, TU Delft

Delft, The Netherlands

Focus on post-hoc interpretability in machine learning.

August 2022

Executive Master in Finance (EMFin), Sharif University of Technology

Tehran, Iran

One-year comprehensive finance programme.

November 2021 – November 2022

Volunteer Activity

- Communications Chair, BNAIC/BeNeLearn 2024 — Joint International Scientific Conferences on AI and Machine Learning, Utrecht University (2024).

Skills

Research Interests: Explainable NLP, large language model evaluation, fairness and cultural alignment in LLMs, annotation reliability, human-AI collaboration, open and reproducible research.

Statistical & ML Methods: Bayesian statistics, reinforcement learning and multi-armed bandits (Thompson sampling), large-scale text analysis (LLMs, transformers, topic modelling, LLM-as-judge), data visualisation, geospatial analysis, high-performance computing, privacy-preserving handling of sensitive data.

Programming & Tools: Python (advanced), R, SQL, Bash; Git and GitHub; PyTorch, TensorFlow, Hugging Face, scikit-learn; LLM application stack (RAG, agents, Model Context Protocol, vector databases); Azure ML, MLflow, Docker, CI/CD; Power BI; LaTeX.

Languages: Farsi (native), English (fluent), Dutch (A2, improving).

References

- Dr. Robert A. Bagheri — Associate Professor, Department of Methodology & Statistics, Utrecht University — a.bagheri@uu.nl
- Dr. Anastasia Giachanou — Assistant Professor, Department of Methodology & Statistics, Utrecht University — a.giachanou@uu.nl
- Prof. Dr. Daniel Oberski — Professor of Social Data Science, Utrecht University; Scientific Director, ODISSEI — d.l.oberski@uu.nl